# Regression-Guided Refocusing Learning With Feature Alignment for Remote Sensing Tiny Object Detection

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***Abstract*— Tiny object detection is a formidable challenge in remote sensing intelligent interpretation. Tiny objects are usually fuzzy, densely distributed, and highly sensitive to positioning errors, which leads to the mainstream detector usually achieving suboptimal detection performance when facing tiny objects. To address the mismatch of mainstream detector architectures and model optimization strategies in the context of tiny object detection, this article presents an efficient and interpretable algorithm for tiny object detection, termed the cross-attention-based feature fusion enhanced tiny object detection network (CAF$^2$ENet). First, the cross-attention mechanism is introduced to refine the upsampling results of deep features. This refinement improves the precision of multiscale feature fusion. Second, a training strategy named regression-based refocusing learning is introduced. Deviating from the conventional optimization strategy, our method guides the optimizer to prioritize higher-quality detection boxes by adjusting sample weights. This adjustment significantly amplifies the detector's potential to achieve superior detection results. Finally, the object composite confidence score is employed for the interpretable filtering of detection boxes. Extensive experiments on tiny object detection in aerial images (AI-TOD) and object detection in optical remote sensing images (DIOR) datasets are carried out, and comparison indicates that the proposed CAF$^2$ENet can perform the remarkable performance compared to other state-of-the-art (SOTA) tiny object detection detectors, as it can reach 63.7% average precision ($AP_{50}$) on AI-TOD and 75.4% $AP_{50}$ on DIOR, achieve SOTA performance.**

***Index Terms*— Feature alignment, loss function, remote sensing, tiny object detection.**

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## I. Introduction

TINY object detection is a classical problem in remote sensing intelligent interpretation with broad application scenarios, including marine surveillance and intelligent transportation [1], [2]. Remote sensing images are often captured from extremely long distances, and due to limitations in sensor resolution, tiny objects are prevalent in these remote sensing images. Compared with the standard size object, the following characteristics of tiny objects pose unique challenges to its detection. As shown in Fig. 1, first of all, the extremely limited pixels of tiny objects make it difficult to extract their features; second, densely arranged tiny objects trial the discrimination ability of the detector. When contrasted with their larger counterparts, tiny objects demonstrate increased sensitivity to errors in bounding box localization. Given these distinctive features of tiny objects, the detector faces increased challenges in accurately classifying and localizing them. Even mainstream detectors [3], [4], [5], [6], [7], [8], which display remarkable performance in object detection tasks within natural scenes, demonstrate suboptimal detection capabilities when confronted with the challenge of identifying tiny objects in remote sensing images. Hence, researching tiny object detection in remote sensing images holds significant value.

In addressing the distinctive characteristics and challenges posed by tiny object detection, various methods have been proposed to enhance the reliability of tasks related to object intelligent interpretation. These methods are specifically crafted to heighten the performance of tiny object detection. Researches into feature extraction and fusion techniques significantly enhance the discriminative capabilities of detectors. Chen et al. [9] incorporate a cross-stage multihead attention module in the detector to further improve the feature discrimination by leveraging the self-attention mechanism. Gong et al. [10] proposed a novel concept, fusion factor, to control information that deep layers deliver to shallow layers, for adapting feature pyramid network (FPN) [11] to tiny object detection. Lim et al. [12] used additional features from different layers as contexts by connecting multiscale features to improve the accuracy of detecting tiny objects. Li et al. [13] proposed the feature layer attention module, which introduces an attention mechanism in the feature layer

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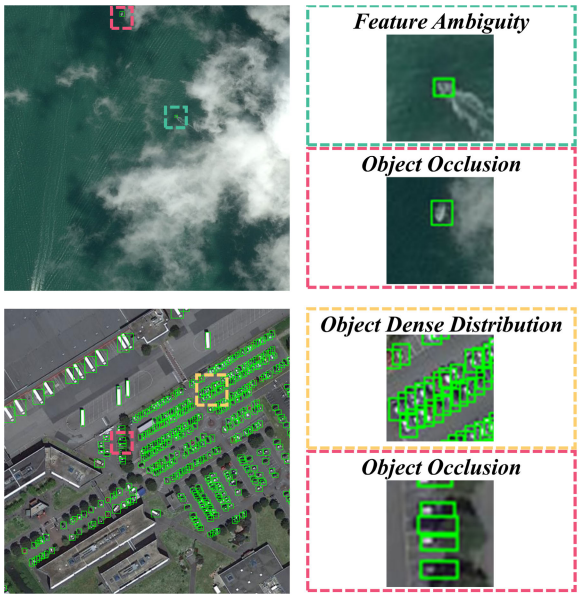



Fig. 1. Tiny objects in remote sensing images typically exhibit the following characteristics: features are blurry, distributed densely, and the background is complex.

to enhance the model's attention to small objects. Researches based on generative adversarial networks (GANs) [14] leverage image super-resolution techniques to enrich the feature information of tiny objects in aerial images, thereby achieving more prominent performance in tiny object detection. The perceptual GANs [15] internally enhance the representation of tiny objects to super-resolution representation to achieve a feature representation similar to that of large objects to improve the discriminative power of the detection model. Inspired by the success of edge-enhanced GAN [16] and enhanced super-resolution GAN [17], Rabbi et al. [18] adopted a novel edge-enhanced super-resolution GAN (EESRGAN) to improve the quality of remote sensing images. In addition, related research that includes substituting non-maximum suppression (NMS) [19] with Soft-NMS [20] can effectively tackle the detection challenges associated with densely arranged objects without necessitating changes to network size or inference time. However, these research efforts typically neglect the global correlation among multiscale features and fall short of fundamentally enhancing the detector's precision in localizing tiny objects. Consequently, these limitations lead to suboptimal detection capabilities for tiny object detectors.

Our motivation is to tackle the challenging issues in classification and regression within the context of tiny object detection. The objective is to improve the detection performance of tiny object detectors by concentrating on both the model structure and model optimization strategy. First, the cross-attention-based upsampling feature refinement (CAFR) module was introduced. Cross-attention is a variant of the self-attention mechanism, excels in establishing correlations between two sequences and thereby fosters associations among multiple input sequences. The CAFR module meticulously refines upsampled deep features, resolving the challenge of feature misalignment inherent in directly employing upsampled deep features for multiscale feature fusion. Concurrently, the cross-attention fosters global correlations among features of different dimensions, empowering the detector to adeptly manage mutual interference between objects and backgrounds. Second, the regression-based refocusing learning was proposed, which dynamically adjusting the learning weights of samples based on their optimization effects. Specifically, under the traditional model optimization strategy, low-quality positive samples with greater losses tend to be optimized first in order to achieve sample balance. However, in the face of tiny objects that are highly sensitive to positioning errors, this optimization strategy cannot guarantee the emergence of effective detection. We abandoned this strategy of sample balance optimization and adopted a training-based dynamic focusing strategy to improve the attention to positive samples with superior positioning effect, and undergo further refinement in subsequent training to enhance positioning accuracy. This embodies a positive feedback adjustment mechanism. Additionally, in order to improve the interpretability of detection box screening, the object composite confidence score was utilized as the basis for post-processing of detection boxes, facilitating the filtering of anchors with exceptional localization effects. The effectiveness of our approach has been validated through extensive experiments conducted on AI-TOD and DIOR, surpassing the current state-of-the-art (SOTA) methods for tiny object detection in aerial images.

The contributions of this article are summarized as follows.

1) An advanced algorithm for tiny object detection, named cross-attention-based feature fusion enhanced tiny object detection network (CAF$^2$ENet), is introduced to achieve highly precise detection of tiny objects. CAF$^2$ENet incorporates a CAFR module and a regression-based refocusing learning. This innovative approach surpasses the performance of current SOTA algorithms for tiny object detection.
2) The CAFR module is developed, which establishes global correlations among multiscale features, realizes upsampling feature refinement, facilitates multiscale feature alignment, and mitigates interference issues among objects and backgrounds.
3) The regression-guided refocusing learning is proposed, which stimulates the potential of the detector to achieve optimal detection performance, enabling high-precision tiny object detection.

## II. Related Works

### A. Tiny Object Detection

As a classic problem in the field of object detection, tiny object detection has always maintained a high research enthusiasm. Addressing the widespread occurrence of tiny objects such as pedestrians and traffic signs in natural scenes, Liu et al. [21] proposed the multiscale region-based convolutional neural network (MR-CNN), which used a multiscale deconvolution operation to upsample the features of the deeper convolution layers and concatenates them to those of the shallow layer to construct the fused feature map.

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Deng et al. [22] proposed an extended feature pyramid network (EFPN), which incorporates a feature texture transfer (FTT) module to super-resolve features and extract credible regional details simultaneously, along with a foreground–background-balanced loss function to alleviate area imbalance of foreground and background. Hu and Dong [23] proposed a feature selection pyramid network (FSPN), which optimizes small target detection through feature reconciliation and spatial context feature selection.

In contrast to the relatively stable contextual information in natural scenes, the background environment in remote sensing images is notably more intricate, featuring a greater degree of various contextual content. Furthermore, the intricate diversity of object categories and the dense distribution of objects in remote sensing images significantly impede the enhancement of the model's ability to detect tiny objects within such imagery. In response to the challenges of detecting tiny objects in remote sensing images, researchers propose countermeasures from multiple perspectives. Chalavadi et al. [24] proposed a network for multiscale object detection in aerial images using hierarchical dilated convolutions (mSODANet), which utilizes the bidirectional feature aggregation module (BFAM) to incorporate dense multiscale contextual features. mSODANet achieved effective multiscale object detection in aerial images. Multibranch feature pyramid network (MB-FPN) [25] used multi-input to form multiscale feature maps for multifeature fusion on different branches, which reduces the formation of pseudo-information and enhances the extraction of detailed features of small objects by the network. Furthermore, the introduction of the global and local combined attention block (GL-CAB), which combines deep global features and shallow local features, addresses the problem of missed detection for small objects. Hong et al. [26] proposed a scale selection pyramid network (SSPNet) for tiny person detection, where context attention module (CAM) takes account of context information to produce hierarchical attention heatmaps, and scale enhancement module (SEM) highlights features of specific scales at different layers, leading the detector to focus on objects of specific scales instead of vast backgrounds. Zhang et al. [27] proposed a new feature pyramid network-based small object detection algorithm, SSRDet, which changes the deep and shallow feature transfer expression of FPN to provide richer contextual information for small object detection. Meanwhile, a lightweight plug-and-play attention module (SPAM) is proposed to reduce the interference of redundant information on small object prediction. Zhang et al. [28] designed a gated context aggregation (GCA) module that selectively aggregates valuable context information to enhance the feature representation of tiny objects, complemented by the development of a positive RoI (pRoI) feature generator in the region convolutional neural network (R-CNN) to generate a rich diversity of high-quality pRoI features for tiny objects. Otherwise, aiming at the difficulty of tiny object label assignment, Xu et al. [29] proposed Gaussian receptive field-based label assignment (RFLA), which utilizes the receptive field distance (RFD) to measure the similarity between the Gaussian receptive field and ground truth (GT), and assigns labels accordingly.

### B. Multiscale Feature Fusion

Conducting research on multiscale feature fusion to achieve more accurate aggregation of contextual information can help improve the performance of object detectors. Wang et al. [30] presented a progressive feature polishing network (PFPN). By employing multiple feature polishing modules (FPMs) in a recurrent manner, PFPN can detect salient objects with fine details without any post-processing. The FPM parallelly updates the features of each level by directly incorporating all higher level context information, while maintaining the dimension and hierarchical structures of the feature maps. Zhang et al. [31] proposed a novel coarse-to-fine framework (CoF-Net) for object detection in remote sensing imagery. In CoF-Net, the fine-grained feature alignment (FFA) module is devised to adaptively learn a fine-grained pixelwise transformation, and establish accurate multiscale feature correspondence. In order to better integrate the features with inconsistent semantics and scales, Dai et al. [32] presented the attention feature fusion (AFF) module to address the problems of contextual aggregation and initial integration. Quan et al. [33] propose a centralized feature pyramid (CFP) for object detection, which is based on a globally explicit centralized feature regulation. Compared to the existing feature pyramids, CFP not only has the ability to capture the global long-range dependencies, but also efficiently obtain an all-round yet discriminative feature representation. Fan et al. [34] proposed a novel multiscale feature fusion module (MSFFM) based on attention mechanism, which utilizes a matrix multiplication operation to measure the relevance between the two feature maps in the spatial dimension. By reweighing feature maps in lower layers, the interference between feature maps in different layers has been reduced.

In tiny object detection tasks, coarse and non-learnable upsampling operations often lead to spatial misalignment when merging multiscale features, resulting in a lack of accurate correspondence between multiscale features. As tiny objects are more sensitive to positioning errors and require higher positioning accuracy, the harm caused by this feature misalignment is more obvious. Therefore, it is necessary to carry out multiscale feature fusion research for tiny object detection.

### C. Model Optimization Strategy

The loss function serves to measure the discrepancy between the model predictions and the truth labels, which is the basis of object detection model optimization and it can guide the optimization direction of the model. Lin et al. [7] proposed the focal loss, which tackles the problem of imbalanced foreground and background samples in the training of one-stage object detectors by introducing a modulating factor into the cross-entropy loss. Yu et al. [35] proposed the utilization of intersection-over-union (IoU) for evaluating bounding box regression quality, creating a unified framework for both evaluation and training metrics. This approach additionally introduces scale invariance to bounding box regression. On this basis, CIoU [36], GIoU [37], and EIoU [38] were put forward one after another. These IoU-based loss functions have been widely adopted across various object detection

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frameworks, leading to significant success. They contribute to the improvement of localization accuracy and robustness in object detectors. Oksuz et al. [39] propose rank sort (RS) loss that defines a ranking objective between positives and negatives as well as a sorting objective to prioritize positives with respect to their continuous IoUs. Due to this ranking-based nature, RS loss can train models in the presence of highly imbalanced data. Ming et al. [40] introduced a matched-sensitive loss to address the challenge of weak correlation between classification and regression, thereby improving the discriminative capacity of high-quality proposals. Class weights are employed to tackle the issue of imbalanced samples among classes, achieved by assigning distinct weights to each class in the loss function.

Traditional object detection model optimization strategies tend to optimize all positive samples for accurate detection. Under this optimization strategy, positive samples with lower detection quality are usually preferentially optimized due to higher losses. This optimization strategy, when applied to detectors for standard-sized objects, facilitates the generation of more effective detection boxes. However, given the heightened sensitivity of tiny objects to localization errors, the outcome of prioritizing low-quality samples is the generation of a substantial number of suboptimal detection boxes. Furthermore, the post-processing approach, relying on the object classification score, lacks adequate interpretability. The inconsistency between classification and regression introduces difficulty in effectively filtering optimal detections. Therefore, it is worthwhile to conduct further research for model optimization strategies.

## III. Proposed Method

In this section, the challenges posed by the feature misalignment issue arising from the direct use of upsampled features for feature aggregation and the problem of ineffective filtering of optimal detections caused by utilizing object classification scores in the post-processing of detection boxes are analyzed. To tackle these difficulties in detecting tiny objects, CAF$^2$ENet is introduced, a novel network architecture designed for tiny object detection, leveraging cross-attention for enhanced feature fusion. Subsequently, detailed insights are provided into two critical components: the CAFR module and the regression-guided refocusing learning.

### A. Overview

The overall framework of CAF$^2$ENet is illustrated in Fig. 2, where the anchor-free detector YOLOX [41] is chosen as the baseline model. Within this framework, the CAFR module is strategically integrated into the neck network to enhance feature fusion. Exploiting the cross-attention mechanism, CAFR establishes global correlations among multiscale features, ensuring precise alignment across various scales. This refinement is crucial for obtaining more accurate feature fusion results within the detector. The regression-guided refocusing learning is introduced in the training phase. This training strategy incorporates a positive feedback mechanism, generating sample weights to guide the optimization of model parameters, consequently, resulting in more accurate object detection results. A detailed description is as follows.

### B. CAFR Module

Mainstream detectors extract multiscale features from input images using a backbone network and subsequently employ an FPN structure to fuse these multiscale features. Features with different scale semantic information are merged through lateral connections and top-down pathways. The original FPN structure directly aggregates deep features with large-scale semantic information with shallow features after upsampling. However, this fusion method fails to establish contextual relevance at a global level and is not effective in enhancing features of tiny objects, especially in complex backgrounds. Additionally, whether using the nearest-neighbor principle or linear interpolation for upsampling, a problem arises with feature misalignment as deep features propagate to shallow features. This feature misalignment will hinder the detector from discriminating dense objects. As shown in Fig. 3, given the higher sensitivity of tiny objects to localization errors, the harm introduced by this coarse feature fusion approach is more pronounced. Establishing global correlations for multiscale features to achieve more accurate upsampled outputs of deep features can effectively alleviate the impact of feature deviation on the detection and localization of densely distributed objects in complex backgrounds. This contributes to improving the performance of tiny object detection.

Inspired by the effectiveness of the self-attention mechanism in establishing global correlations within input sequences, the CAFR module was introduced, The structure of CAFR is illustrated in Fig. 2, and the structure of cross-attention is presented in Fig. 4. CAFR fine-tunes the upsampled deep features by establishing global correlations among multiscale features. Its integration alleviates the feature misalignment issues arising from the direct use of upsampled deep features, consequently bolstering the detector's discriminative capabilities for dense tiny objects within complex backgrounds. CAFR takes two adjacent feature maps of different scales as input and produces refined upsampled features. In this process, CAFR initiates the establishment of global correlations between the two scale feature maps through the cross-attention mechanism. Details of the calculation can be found in the following equation:

$$\begin{aligned}&\text{Cross Attention}(P_{i+1}, C_i)\\&= \text{Softmax}\left(\frac{\left(PE_{i+1}(Up(P_{i+1}))W_q\right)(PE_i(C_i)W_k)^T}{\sqrt{d_k}}\right)\\&\quad\times (PE_i(C_i)W_v)\end{aligned} \tag{1}$$

where $C_i \in \mathbb{R}^{C\times H_i\times W_i}$ is a non-deepest feature map output by the backbone network, $P_{i+1} \in \mathbb{R}^{C\times H_{i+1}\times W_{i+1}}$ is the deeper feature map adjacent to $C_i$ after FPN processing, and $H_i = 2H_{i+1}$, $W_i = 2W_{i+1}$. $Up(*)$ refers to the upsampling of feature maps. $PE(*)$ can expand the input feature map into a feature sequence. $W_q, W_k, W_v \in \mathbb{R}^{H_iW_i\times C}$ are learnable linear transformations that map the input feature sequence into

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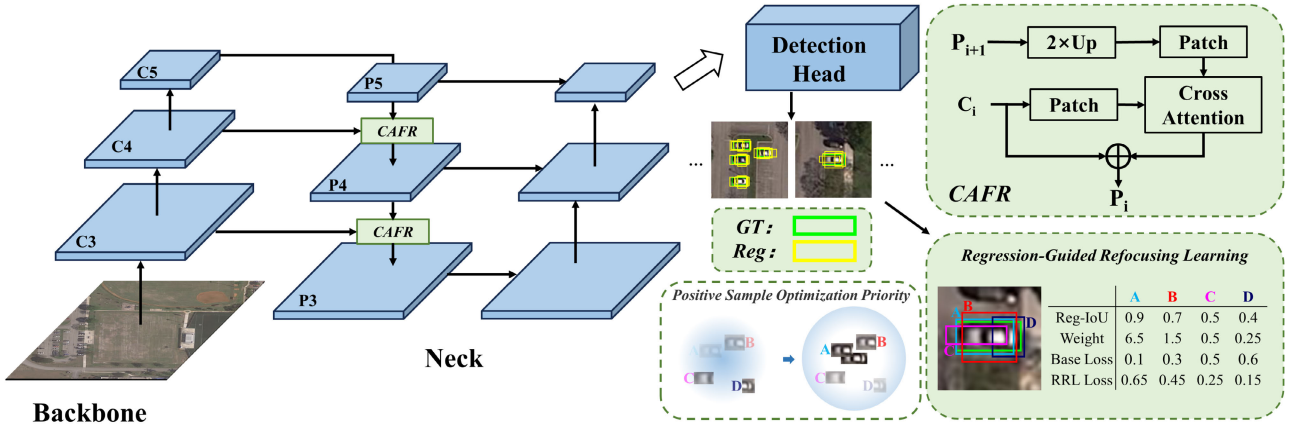



Fig. 2. Overall structure of CAF$^2$ENet. CAF$^2$ENet architecture takes anchor-free detector YOLOX as the baseline, CSPDarkNet53 as the backbone to extract multiscale features from the input aerial image, FPN and PAN structures are combined as the neck for feature fusion, and the detection results are output by the detection head. CAFR module is based on cross-attention mechanism and is used to refine the upsampled deep feature map to align it with its adjacent shallow feature in the spatial dimension. The loss function of CAF$^2$ENet is weighted by the regression-guided refocusing learning. The optimization priorities of the samples are rearranged.

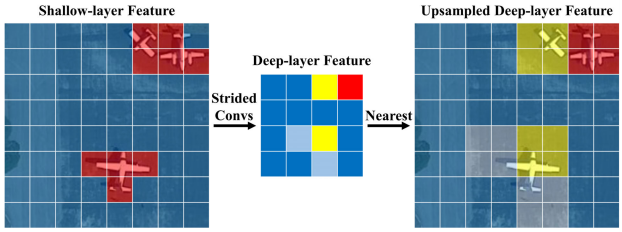



Fig. 3. Problem illustration. Simply upsampling the deep features cannot recover the original spatial correspondence and will cause feature misalignment during fusing different levels of features.

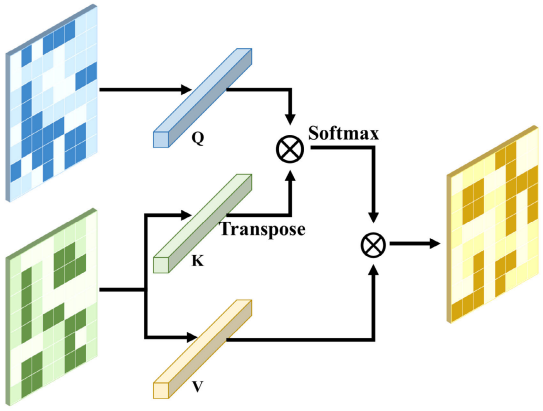



Fig. 4. Illustration of cross-attention mechanism, where $Q$ is transformed from the upsampled deep feature map, and $K$, $V$ are transformed from the shallow feature map.

query $Q$, key $K$, and value $V$. $(*)^T$ represents the transpose operation on matrices and $d_k$ represents the length of each vector in the input sequence.

After computing the cross-attention across adjacent feature maps, the resulting calculations are directly integrated with the upsampled deep features, thereby yielding a refined version of the upsampled deep features. The specific computations are detailed in the following equation:

$$\text{Refine}(P_{i+1}, C_i) = Up(P_{i+1}) + \text{Cross Attention}(P_{i+1}, C_i). \tag{2}$$

CAFR utilizing cross-attention refines the upsampled deep features, aligning multiscale features. The resulting refined features alleviate the interference caused by feature misalignment in the accurate localization of dense objects. Additionally, the establishment of global correlations among multiscale features concurrently mitigates the impact of complex backgrounds on object detection, ultimately enhancing the performance of the tiny object detector.

### C. Regression-Guided Refocusing Learning

The object detection model generates detection results through the detection head after extracting and fusing features from the input images. Following this, duplicate detection boxes are eliminated using non-maximum suppression based on the object classification scores. However, the misalignment between object classification scores and object localization accuracy is widespread, and post-processing steps reliant on object classification scores frequently lack ample interpretability. As the detection boxes with superior localization accuracy are restrained due to lower object classification scores, the detector is prevented from achieving the optimal performance. Simultaneously, as depicted in Fig. 5, tiny objects exhibit heightened sensitivity to localization errors. Under equivalent absolute error conditions, tiny objects demonstrate more pronounced variations in IoU. Therefore, tiny object detection requires higher positioning accuracy, and the impact arising from the inconsistency between classification and regression will be further amplified.

The influence of object classification and regression inconsistency on tiny object detection is demonstrated. As shown in Fig. 6, the image in the upper-right corner illustrates a scenario where a low-quality detection box suppresses a

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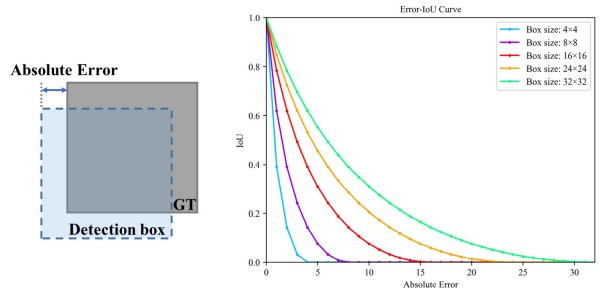



Fig. 5. Problem illustration. Tiny objects are more sensitive to positioning errors. Under the condition of the same absolute error, the IoU of tiny object changes more dramatically.

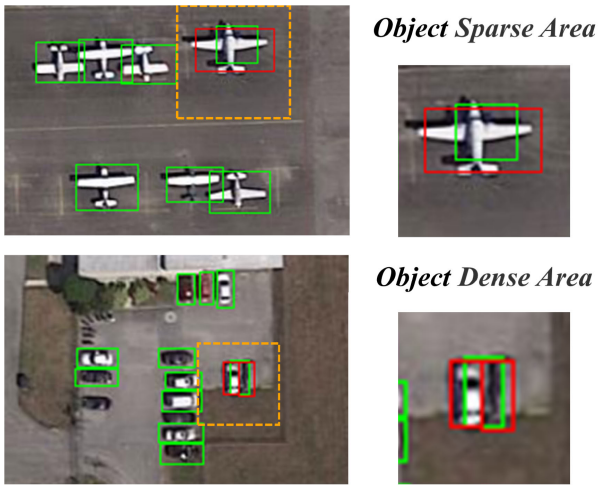



Fig. 6. Remote sensing tiny object detection results show a weak correlation between positioning accuracy and classification confidence, which is not conducive to selecting accurate detection results through classification scores in the post-processing process.

high-quality detection box in sparse areas of objects. Clearly, detections with higher quality (red boxes) are inhibited by low-quality detections (green boxes) due to their relatively lower classification scores. The lower-right image illustrates a scenario in densely object areas where detection boxes with poor localization accuracy (green boxes) simultaneously suppress multiple more accurate detection boxes (red boxes), leading to missed detections.

Conventional research on weighted loss functions primarily aims to enhance the discrimination between objects and backgrounds, enlarging the differences between classes. However, there is limited exploration on adjusting weights for various positive samples within a single GT to increase the likelihood of obtaining higher-quality detection boxes. To stimulate the potential of the detector in generating precise detection boxes, especially for tiny objects, and simultaneously improve the interpretability of object detection box filtering, the regression-guided refocusing learning (RRL) approach was proposed. This training strategy abandons the optimization method of giving priority to low quality and instead utilizes a positive feedback regulation mechanism to increase the weight of high-quality regression box loss during training. This encourages the detector to obtain a minimal yet optimal set of detections. In addition, the introduction of object localization confidence scores based on regression accuracy enhances interpretability during the post-processing stage of detection box filtering. Further details of our loss function weighting paradigm are elaborated below. For each sample of GT, the weight is defined by the following equation:

$$w_{g,p} = 1 + \alpha(t) \times \left( \frac{1 - 2 \times \mathrm{IoU}_{\mathrm{mean}} + \mathrm{IoU}_{g,p}}{1 - \mathrm{IoU}_{g,p}} - 1 \right) \tag{3}$$

where $g$ is a GT of the input image, $p$ represents a positive sample belonging to $g$, IoU is the set of IoU between the regression boxes of all positive samples belonging to $g$, and $g$, $\mathrm{IoU}_{g,p}$ represents the IoU between the regression box of $p$ and $g$, and $\mathrm{IoU}_{\mathrm{mean}}$ denotes the average of IoU. For training stability, a modulation parameter $\alpha(t)$ was employed based on the training process. In the initial stages of training, the additional positive feedback weight for all samples is set to 1. As training iterations progress, the impact of the positive feedback adjustment weight on the loss function gradually increases. The adjustment of $\alpha(t)$ at each epoch is based on the following equation:

$$\alpha(t) = \begin{cases} 0, & 0 \le t < 0.25 \\ \dfrac{t - 0.25}{0.5 - 0.25}, & 0.25 \le t < 0.5 \\ 1, & t \ge 0.5 \end{cases} \tag{4}$$

where $t$ = (epoch/epochs), epochs is the total number of training epochs, and epoch is the current training epoch. Under the adjustment of our regression-guided refocusing learning, the weight of positive samples with better regression effect is increased, on the contrary, the weight of positive samples with poor regression effect is reduced. Variations in the prioritization of positive sample optimization are presented, as shown in Fig. 2. Specifically, the clarity of the detection box is positively related to the attention of the detection model. The regression-guided refocusing learning guides the detector to pay more attention to positive samples with excellent regression performance, promoting better optimization in subsequent training. The entire process of regression-guided refocusing learning is detailed in Algorithm 1.

On the basis of our training strategy, in order to ensure that high-quality detection boxes can be correctly selected in the post-processing stage, the object positioning confidence score based on regression is proposed. The object positioning confidence score is output from the object regression branch, and the binary cross-entropy loss is used to optimize its value. The specific loss is defined as the following equation:

$$L_{\mathrm{pos}} = \frac{1}{N_p} \left( \sum_{g \in \varphi_g} \sum_{p \in \varphi_{p_g}} w_{g,p} \cdot \mathrm{Loss}_{\mathrm{BCE}}(o_p, o_{\mathrm{IoU}}) + \sum_{n \in \varphi_n} w_n \cdot \mathrm{Loss}_{\mathrm{BCE}}(o_n, 0) \right) \tag{5}$$

where $\varphi_g$ is the set of all GT, $\varphi_{p_g}$ is the set of positive samples assigned to $g$, $\varphi_n$ is the set of negative samples, $N_p$ is the total number of positive samples, and $o_p$ and $o_n$ represent the object confidence scores of positive and negative samples output from the detection head. $o_{\mathrm{IoU}}$ represents the IoU of the decoded $p$ and $\varphi_g$. The weight $w_n$ of negative samples is 1.0 by default,

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**Algorithm 1** Regression-Guided Refocusing Learning

**Input:**

$\mathcal{G}$ is a set of ground-truth boxes on the image
*epochs* is the total number of training iterations
$\lambda$ is the predefined hyper-parameter

**Output:**

$L_{total}$ represents the total loss of the detector

**for** $epoch = 1, 2, 3, \ldots, epochs$ **do**
  $t = \frac{epoch}{epochs}$
  **for** $g \in \mathcal{G}$ **do**
    Assign positive samples to $g$ according to the label assignment strategy
    $\mathcal{P}_g$ is the set of positive samples belonging to $g$
    Calculate the $IoU_{mean}$ of $\mathcal{P}_g$
    **for** $p \in \mathcal{P}_g$ **do**
      Calculate the weight of $p$ according to Eq. 3, where $\alpha(t)$ is expressed by Eq. 4
    **end for**
  **end for**
  Calculate the object positioning confidence loss $L_{pos}$ using Eq. 5
  Calculate the regression loss $L_{reg}$ using Eq. 6
  Calculate the classification loss $L_{cls}$ using Eq. 7
  $L_{total} = L_{cls} + \lambda L_{reg} + L_{pos}$
**end for**
**return** $L_{total}$

and the weight $w_{g,p}$ of positive samples is calculated by 3. The introduction of object confidence score makes the selection of detection box interpretable.

In addition, the regression-guided refocusing learning loss is also utilized in calculating both regression and classification losses. The regression loss is computed using the IoU loss, while the binary cross-entropy loss is employed for determining the classification loss. Note that the aforementioned two losses are only compute for positive samples, with the specific formulas detailed in the following equation:

$$L_{\text{reg}} = \frac{1}{N_p} \sum_{g \in \varphi_g} \sum_{p \in \varphi_{p_g}} w_{g,p} \cdot \text{Loss}_{\text{IoU}}(r_p, gt_g) \tag{6}$$

$$L_{\text{cls}} = \frac{1}{N_p} \sum_{g \in \varphi_g} \sum_{p \in \varphi_{p_g}} w_{g,p} \cdot \text{Loss}_{\text{BCE}}(c_p, s_p). \tag{7}$$

The definition of the same variable is referenced to 5. In the regression loss, $r_p$ denotes the regression box for $p$, while $gt_g$ represents the bounding box of $g$. $\text{Loss}_{\text{IoU}}$ signifies the IoU loss, with its specific calculation detailed in 8, where $\text{IoU}(*, *)$ denotes the calculation of the IoU between two bounding boxes. In the classification loss, $c_p$ denotes the object classification output for positive samples. $s_p$ represents the expected object classification result of $p$. $\text{Loss}_{\text{BCE}}$ represents binary cross-entropy loss, as defined by the following equation:

$$\text{Loss}_{\text{IoU}}(r, gt) = 1 - \text{IoU}(r, gt) \tag{8}$$

$$\text{Loss}_{\text{BCE}}(x, y) = -\sum_{k=1}^{N_c} (y_k \ln x_k + (1 - y_k) \ln(1 - x_k)). \tag{9}$$

The total loss of the detector is calculated as follows, where $\lambda$ is the balancing coefficient of regression loss, with a default value of 5.0

$$L_{\text{total}} = L_{\text{cls}} + \lambda L_{\text{reg}} + L_{\text{pos}}. \tag{10}$$

In the post-processing stage of the detector, the object composite positioning classification score $S_{\text{obj}}$ is used to determine the retention and suppression of the detection box, and its calculation method is shown in the following equation:

$$S_{\text{obj}} = S_{\text{pos}} \times S_{\text{cls}} \tag{11}$$

where $S_{\text{pos}}$ and $S_{\text{cls}}$ are the object positioning score and classification score output by the detector, respectively. The regression-guided refocusing learning inspires the potential of the tiny object detector to obtain higher-quality detection boxes. Simultaneously, the box filtering method based on classification scores and object positioning confidence is more reliable and interpretable.

## IV. Experiments

### A. Dataset

AI-TOD [42] benchmark is a remote sensing dataset designed for tiny object detection, which was established by Wuhan University based on public large aerial image datasets. It comprises 280 036 aerial images of size $800 \times 800$ pixels, with a total of 700 621 annotated instances. AI-TOD includes eight object classes: airplane (AI), bridge (BR), storage-tank (ST), ship (SH), swimming-pool (SP), vehicle (VE), person (PE), and wind-mill (WM). The average size of the objects in AI-TOD is only 12.8 pixels, much smaller than that of typical remote sensing datasets like DOTA [43] (55.2 pixels). A training set of 11 214 images and a validation set of 2804 images, totaling 14 018 images, were employed for training, while the test set, consisting of 14 018 images, was used to evaluate the performance of the detector.

DIOR [44] is a large-scale benchmark dataset for optical remote sensing image object detection, open-sourced by Northwestern Polytechnical University in 2019. The dataset covers 20 object classes, and comprises 23 463 remote sensing images and 190 288 annotated object instances marked with axis-aligned bounding boxes. The images in the DIOR dataset have a size of $800 \times 800$ pixels, and spatial resolutions ranging from 0.5 to 30 m.

### B. Evaluation Metrics

In the experiment, the average precision (AP) is used to evaluate the detection performance of the model, which is defined as follows:

$$\text{AP} = \int_0^1 P(r) dr \tag{12}$$

where $r$ is the Recall rate and $P(r)$ is the corresponding Precision. The Recall and Precision are calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

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TABLE I
EVALUATION RESULTS OF DIFFERENT MODELS ON AI-TOD DATASET

| Method | Backbone | AI | BR | ST | SH | SP | VE | PE | WM | $AP$ | $AP_{50}$ | $AP_{75}$ | $AP_{vt}$ | $AP_t$ | $AP_s$ | $AP_m$ | FPS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| TridentNet [8] | ResNet-50 | 7.6 | 0.0 | 10.8 | 9.6 | 2.5 | 7.0 | 1.9 | 0.0 | 4.9 | 13.1 | 2.7 | 0.0 | 2.8 | 10.8 | 19.2 | 7.6 |
| Faster R-CNN [3] | ResNet-50 | 19.6 | 1.4 | 17.3 | 17.4 | 7.9 | 10.4 | 3.6 | 0.0 | 9.7 | 22.1 | 6.8 | 0.0 | 5.2 | 21.7 | 32.5 | 25.6 |
| Mask R-CNN [4] | ResNet-50 | 20.0 | 1.9 | 17.4 | 18.6 | 8.0 | 10.9 | 3.8 | 0.0 | 10.1 | 22.7 | 7.6 | 0.0 | 5.8 | 22.1 | 32.6 | 14.0 |
| Cascade R-CNN [5] | ResNet-50 | 21.0 | 6.7 | 20.3 | 21.0 | 7.7 | 12.8 | 4.8 | 0.0 | 11.8 | 25.6 | 9.5 | 0.1 | 7.9 | 23.0 | 34.8 | 14.6 |
| Dynamic R-CNN [50] | ResNet-50 | 10.4 | 15.5 | 30.7 | 38.7 | 7.0 | 19.5 | 9.4 | 3.9 | 16.9 | 39.4 | 11.6 | 6.3 | 18.2 | 21.2 | 25.6 | 17.1 |
| PAFPN [51] | ResNet-50 | 19.8 | 1.4 | 17.0 | 17.5 | 7.7 | 10.4 | 3.7 | 0.0 | 9.7 | 22.2 | 6.8 | 0.0 | 5.4 | 21.6 | 32.5 | 16.6 |
| DetectoRS [52] | ResNet-50 | 25.3 | 8.2 | 22.9 | 25.4 | 13.2 | 15.2 | 5.9 | 0.1 | 14.5 | 32.2 | 10.9 | 0.1 | 10.6 | 27.7 | 37.7 | 5.0 |
| RetineNet [7] | ResNet-50 | 0.4 | 0.0 | 2.4 | 10.0 | 0.0 | 5.8 | 1.0 | 0.0 | 2.4 | 8.4 | 0.7 | 0.8 | 2.7 | 4.0 | 8.8 | 17.6 |
| SSD-512 [6] | VGG-16 | 10.9 | 2.1 | 10.7 | 14.3 | 1.8 | 9.2 | 2.2 | 0.9 | 6.5 | 21.0 | 2.4 | 1.0 | 5.3 | 11.9 | 21.2 | 29.9 |
| YOLOv3 [48] | DarkNet-53 | 19.1 | 9.7 | 21.7 | 20.9 | 4.7 | 14.5 | 5.2 | 2.7 | 12.3 | 35.3 | 5.6 | 3.2 | 12.0 | 18.5 | 24.2 | 23.5 |
| TOOD [53] | ResNet-50 | 22.2 | 5.3 | 28.9 | 27.9 | 9.1 | 18.6 | 7.0 | 0.5 | 14.9 | 34.7 | 10.6 | 3.3 | 12.8 | 21.7 | 33.1 | 12.3 |
| ATSS [54] | ResNet-50 | 0.5 | 9.8 | 16.8 | 26.4 | 0.6 | 12.0 | 4.0 | 1.2 | 8.9 | 22.5 | 5.3 | 2.1 | 10.1 | 11.4 | 15.0 | 16.3 |
| AutoAssign [55] | ResNet-50 | 14.7 | 10.7 | 25.5 | 26.4 | 4.1 | 16.5 | 6.0 | 3.0 | 13.4 | 36.2 | 6.9 | 3.5 | 13.5 | 17.0 | 24.9 | 14.9 |
| FCOS [56] | ResNet-50 | 17.2 | 1.6 | 21.2 | 19.8 | 0.8 | 13.3 | 4.9 | 0.0 | 9.8 | 24.1 | 6.2 | 1.2 | 7.8 | 16.1 | 27.2 | 11.6 |
| RepPoints [57] | ResNet-50 | 0.0 | 0.0 | 12.7 | 16.1 | 0.0 | 10.6 | 2.4 | 0.0 | 5.2 | 15.2 | 2.2 | 1.3 | 5.4 | 7.5 | 13.7 | 14.0 |
| FoveaBox [58] | ResNet-50 | 14.7 | 0.0 | 19.2 | 17.7 | 0.0 | 12.4 | 4.2 | 0.0 | 8.5 | 20.6 | 5.7 | 0.9 | 6.1 | 14.1 | 27.2 | 20.0 |
| CenterNet [59] | ResNet-18 | 12.8 | 2.0 | 15.3 | 15.9 | 1.7 | 11.7 | 4.6 | 1.5 | 8.2 | 24.3 | 3.4 | 1.3 | 6.9 | 12.9 | 22.3 | 29.7 |
| M-CenterNet [42] | DLA-34 | 18.6 | 10.6 | 27.6 | 22.3 | 7.5 | 18.6 | 9.2 | 2.0 | 14.5 | 40.7 | 6.4 | 6.1 | 15.0 | 19.4 | 20.4 | - |
| FSANet [60] | Swin-T | 30.9 | 15.1 | 35.0 | 40.3 | 19.8 | 24.9 | 8.9 | 5.6 | 22.6 | 52.8 | 15.6 | 7.4 | 21.6 | 29.1 | 38.5 | - |
| DetectoRS w/RFLA [29] | ResNet-50 | 27.6 | 17.1 | 36.8 | 44.8 | 15.5 | 24.7 | 11.3 | 5.8 | 23.0 | 52.8 | 16.4 | 8.8 | 23.1 | 28.3 | 36.9 | 5.0 |
| MENet [28] | Swin-T | 27.5 | 16.4 | 37.4 | 42.1 | 18.8 | 24.9 | 10.2 | 8.2 | 23.2 | 56.2 | 15.0 | 9.7 | 23.9 | 25.3 | 34.4 | - |
| YOLOX-N [41] | DarkNet-53 | 15.0 | 2.2 | 26.2 | 20.6 | 2.4 | 16.2 | 5.6 | 0.4 | 11.1 | 29.1 | 5.9 | 4.2 | 10.6 | 15.7 | 16.8 | **42.9** |
| YOLOX-T | DarkNet-53 | 23.3 | 6.5 | 34.5 | 27.5 | 4.7 | 25.0 | 8.3 | 1.6 | 16.4 | 40.9 | 10.0 | 6.0 | 15.7 | 22.2 | 23.4 | 40.2 |
| YOLOX-S | DarkNet-53 | 30.7 | 12.2 | 36.8 | 32.3 | 10.8 | 28.3 | 11.0 | 2.4 | 20.6 | 49.4 | 13.8 | 7.3 | 19.5 | 28.0 | 33.8 | 36.7 |
| YOLOX-M | DarkNet-53 | 34.3 | 15.8 | 38.7 | 35.2 | 15.9 | 30.3 | 12.9 | 4.6 | 23.4 | 54.1 | 16.5 | 9.5 | 22.5 | 31.6 | 36.7 | 34.0 |
| **CAF²ENet-N** | DarkNet-53 | 29.0 | 12.9 | 35.2 | 34.6 | 8.3 | 27.2 | 10.4 | 3.7 | 20.2 | 48.1 | 13.3 | 6.0 | 19.9 | 26.5 | 29.7 | 42.0 |
| **CAF²ENet-T** | DarkNet-53 | 33.2 | 16.8 | 40.4 | 42.8 | 21.3 | 33.3 | 14.1 | 4.5 | 25.8 | 57.3 | 19.9 | 10.8 | 26.0 | 32.6 | 36.0 | 37.1 |
| **CAF²ENet-S** | DarkNet-53 | 38.5 | 18.2 | 41.5 | 45.8 | 22.9 | 34.5 | 15.8 | 7.7 | 28.1 | 61.4 | 21.8 | 12.4 | 27.9 | 34.9 | 39.2 | 34.3 |
| **CAF²ENet-M** | DarkNet-53 | **39.7** | **22.8** | **43.7** | **50.6** | **24.6** | **34.9** | **17.1** | **8.9** | **30.2** | **63.7** | **25.0** | **12.8** | **30.3** | **36.7** | **41.9** | 31.6 |

The best indicators are highlighted in bold, and the suboptimal indicators are underlined.

where true positive (TP), FP, and FN represent the numbers of TP samples, false positive (FP) samples, and false negative (FN) samples defined under different IoU matching case, respectively. MS COCO [45] AP metrics are utilized to evaluate the performance of various detectors on AI-TOD. The detailed description is, $AP_{50}$ means the IoU threshold of defining TP is 0.5, $AP_{75}$ means the IoU threshold of defining TP is 0.75, AP means the average value from $AP_{50}$ to $AP_{95}$, with an IoU interval of 0.05. $AP_{vt}$, $AP_t$, $AP_s$, and $AP_m$ correspond to AP values distinguished based on object size. Specifically, $AP_{vt}$ represents the AP for objects with a size less than $8^2$, $AP_t$ for objects with sizes between $(8^2, 16^2]$, $AP_s$ for objects with sizes between $(16^2, 32^2]$, and $AP_m$ for objects with sizes greater than $32^2$.

To assess the processing efficiency of the object detector, the parameters (Params) and frames per second (FPS) metrics are introduced. The Params metric represents the total number of parameters in the detector that need to be trained, with more parameters indicating greater storage space required, measured in Megabits (MB). FPS denotes the number of images processed per second and is used to evaluate the efficiency of the tiny object detector. Generally, a higher FPS indicates better processing efficiency of the detector.

### C. Implementation Details

Our experimental code is based on the Pytorch and mmdetection [46] toolkit. Unless specified otherwise, the pre-trained model of backbone is based on the ImageNet [47] dataset. YOLOX [41] is an improved anchor-free detector based on YOLOv3 [48]. The adoption of the SimOTA label assignment strategy and the introduction of decoupled heads enhance the detection accuracy of YOLOX and expedite the model convergence speed. According to the different complexity and storage requirements, YOLOX encompasses different variants such as YOLOX-N, YOLOX-T, YOLOX-S, and YOLOX-M. CAF²ENet follows the YOLOX architecture, and endnotes are used to mark their corresponding baselines. We optimize the CAF²ENet model using the stochastic gradient descent (SGD) [49] optimizer, with an initial learning rate set to 0.01 and a momentum of 0.9. The total number of epochs is set to 300. The experiments were conducted on an NVIDIA GeForce RTX 3090 GPU, with a batch size of 16.

### D. Comparative Experiments

We compare CAF²ENet with other most advanced object detection models on AI-TOD and DIOR datasets, including some remote sensing tiny object detection algorithms that achieve SOTA performance.

*1) Main Results on AI-TOD:* Table I shows the experimental results of different detectors in the same training environment on AI-TOD. Obviously, compared with other methods, our proposed method achieves the highest accuracy of remote sensing tiny object detection. First, our method is directly dependent on the performance of the baseline model, and several versions of YOLOX family are used by us to carry out experiments. Taking YOLOX-N, YOLOX-T, YOLOX-S, and YOLOX-M as baselines, respectively, we have built CAF²ENet-N, CAF²ENet-T, CAF²ENet-S, and CAF²ENet-M with different complexity, and compared with their respective baselines, the $AP_{50}$ metrics have been improved by 19.0%,

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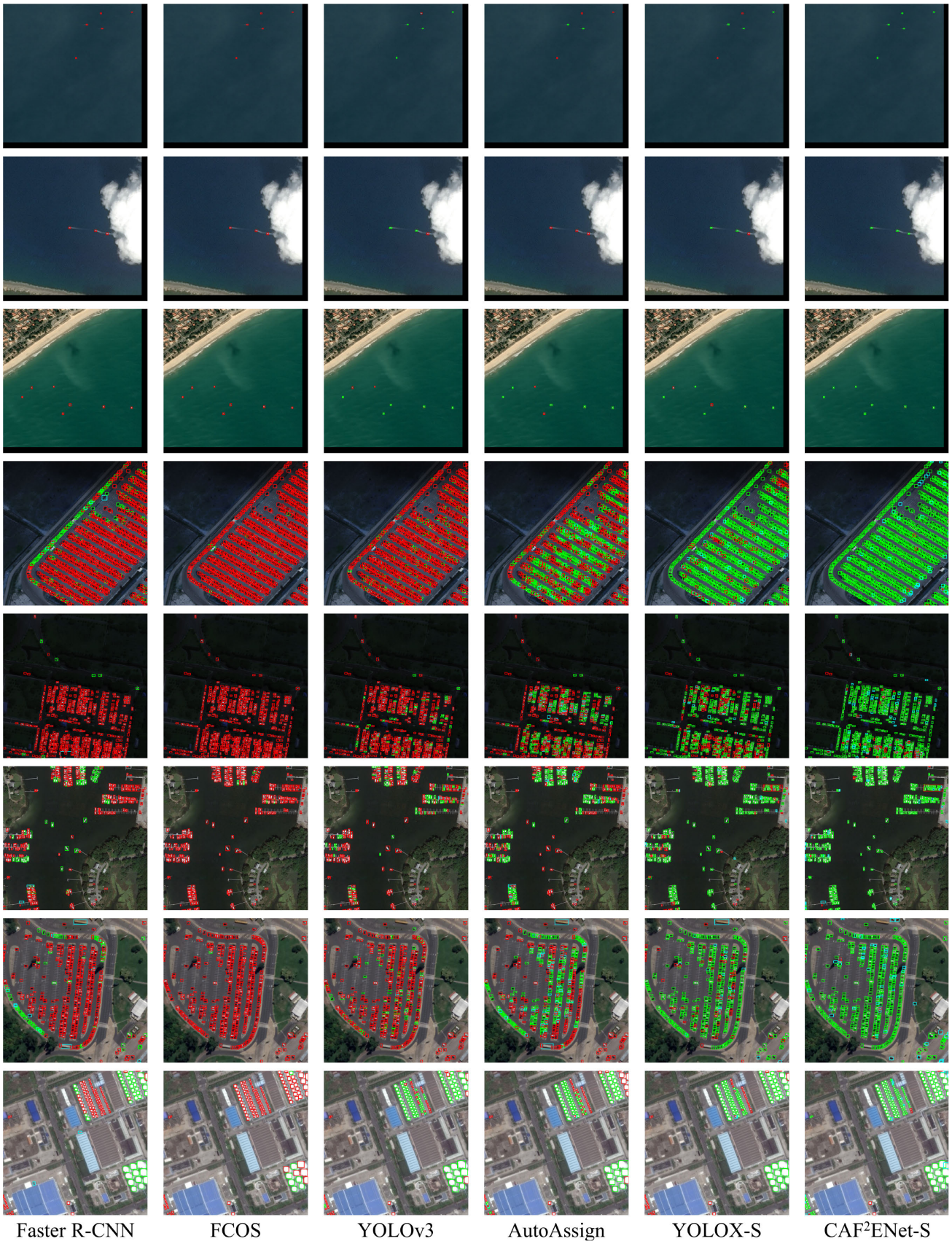



Fig. 7. Visualization of detection results of different object detection models on AI-TOD dataset. The green boxes represent the TP predictions. The cyan boxes represent the FP predictions. The red boxes represent the FN predictions.

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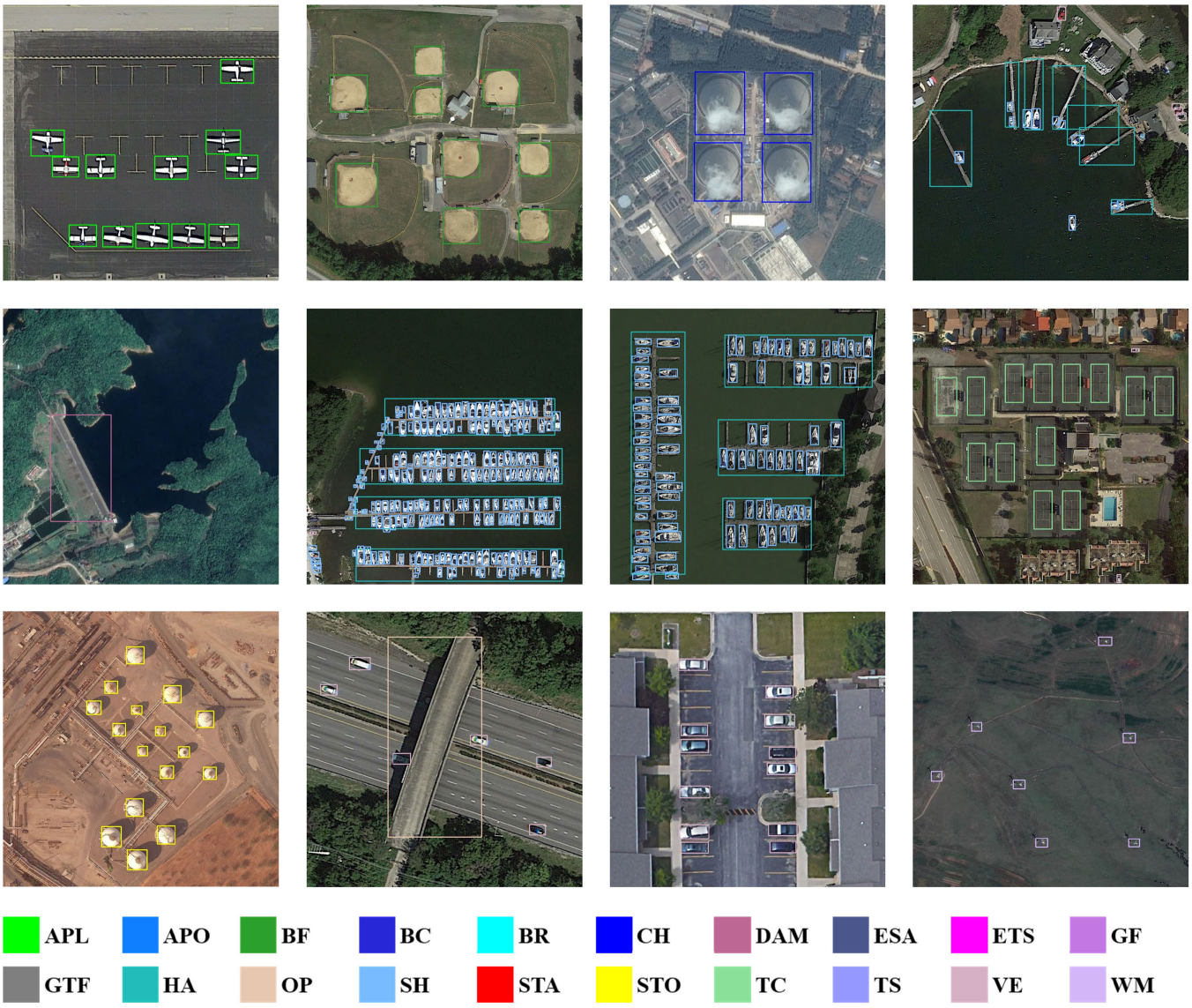



Fig. 8. Visualization of detection results by CAF$^2$ENet on the DIOR dataset. Detection boxes for different object categories are annotated using different colors.

16.4%, 12.0%, and 9.6%, respectively, and the other evaluation metrics are also improved. Second, compared with the traditional object detection network, our detection performance is greatly ahead, and at the same FPS, our method achieves optimal performance in tiny object detection. Additionally, compared with other tiny object detection algorithms that achieved SOTA performance on AI-TOD, our method demonstrates superior performance. Our best $AP_{50}$ metric surpasses that of M-CeneterNet [42], DetectoRS w/RFLA [29], FSANet [60], and MENet [28] by 23.0%, 10.9%, 10.9%, and 7.5%, respectively, accompanied by improvements in other evaluation metrics. The experimental results unequivocally demonstrate that our model achieves SOTA performance on AI-TOD. Fig. 7 showcases a selection of our detecting results on the test set.

*2) Main Results on DIOR:* To validate the versatility of our method, we conducted comparative experiments on the DIOR benchmark, and the results are presented in Table II. The visualization of detection results by CAF$^2$ENet-M on the DIOR dataset is shown in Fig. 8, where detection boxes annotated with different colors are used to indicate objects of various classes. The comparative experiments on the DIOR dataset demonstrate that our algorithm exhibits excellent generalization ability. First, in comparison to the baseline model, our approach notably enhances the $AP_{vt}$, $AP_t$, and $AP_s$ metrics, indicating superior overall detection performance. Additionally, when contrasted with the tiny object detection algorithm DetectoRS w/RFLA [29], our algorithm consistently exhibits superior detection proficiency across a spectrum of object sizes. Moreover, the best $AP_{50}$ metric of our method reaches 75.4%, which surpasses current mainstream detectors and reaches the SOTA level on DIOR benchmark.

### *E. Ablation Experiments*

*1) Evaluation of Different Weighting Methods:* The influence of different calculation methods of loss function weights

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TABLE II
EVALUATION RESULTS OF DIFFERENT MODELS ON DIOR DATASET

| Method | $AP$ | $AP_{50}$ | $AP_{75}$ | $AP_{vt}$ | $AP_t$ | $AP_s$ | $AP_m$ |
|---|---|---|---|---|---|---|---|
| TridentNet | 31.1 | 53.4 | 31.9 | 0.0 | 0.3 | 3.9 | 40.2 |
| Faster R-CNN | 32.3 | 55.8 | 32.7 | 0.0 | 1.4 | 9.6 | 41.1 |
| Cascade R-CNN | 36.2 | 57.5 | 38.7 | 0.0 | 1.7 | 10.2 | 45.6 |
| Dynamic R-CNN | 34.2 | 55.7 | 35.9 | 0.0 | 1.5 | 9.8 | 43.3 |
| PAFPN | 33.5 | 57.2 | 33.9 | 0.0 | 1.4 | 9.6 | 42.4 |
| DetectoRS | 41.9 | 63.8 | 45.7 | 0.0 | 2.0 | 11.5 | 51.8 |
| RetinaNet | 18.0 | 35.5 | 15.9 | 0.0 | 0.0 | 1.5 | 25.2 |
| SSD-512 | 33.4 | 60.2 | 33.4 | 0.1 | 0.7 | 8.3 | 41.8 |
| YOLOv3 | 40.9 | 64.3 | 43.7 | 0.1 | 3.3 | 12.4 | 50.2 |
| TOOD | 38.4 | 60.3 | 40.9 | 0.0 | 1.9 | 10.3 | 48.0 |
| ATSS | 15.3 | 36.6 | 10.2 | 0.0 | 1.0 | 2.9 | 20.1 |
| AutoAssign | 37.8 | 63.3 | 38.7 | 0.2 | 3.0 | 11.4 | 46.5 |
| FCOS | 32.8 | 57.2 | 32.7 | 0.0 | 1.0 | 8.1 | 41.8 |
| RepPoints | 27.3 | 45.1 | 28.0 | 0.0 | 0.8 | 6.8 | 35.9 |
| FoveaBox | 31.1 | 51.8 | 32.0 | 0.0 | 1.1 | 8.1 | 40.2 |
| CenterNet | 33.7 | 54.6 | 35.3 | 0.0 | 0.8 | 6.4 | 43.2 |
| DetectoRS w/RFLA | 46.5 | 68.1 | 50.8 | 0.3 | 5.1 | 14.5 | 55.7 |
| YOLOX-N | 30.1 | 52.5 | 30.4 | 0.4 | 4.1 | 13.9 | 36.5 |
| YOLOX-T | 36.8 | 60.1 | 38.7 | 0.7 | 6.8 | 17.3 | 44.0 |
| YOLOX-S | 45.5 | 70.2 | 49.1 | 1.1 | 7.9 | 19.7 | 53.5 |
| YOLOX-M | 48.6 | 72.5 | 52.9 | 1.1 | 8.5 | 20.9 | 56.7 |
| **CAF²ENet-N** | 38.7 | 63.3 | 40.3 | 0.5 | 4.1 | 12.0 | 47.2 |
| **CAF²ENet-T** | 42.3 | 65.7 | 46.0 | 0.9 | 5.6 | 14.3 | 51.0 |
| **CAF²ENet-S** | 48.6 | 72.1 | 52.8 | **1.6** | 8.0 | 18.8 | 57.2 |
| **CAF²ENet-M** | **53.4** | **75.4** | **58.7** | 1.4 | **8.6** | **21.0** | **62.1** |

The best indicators are highlighted in bold, and the suboptimal indicators are underlined.

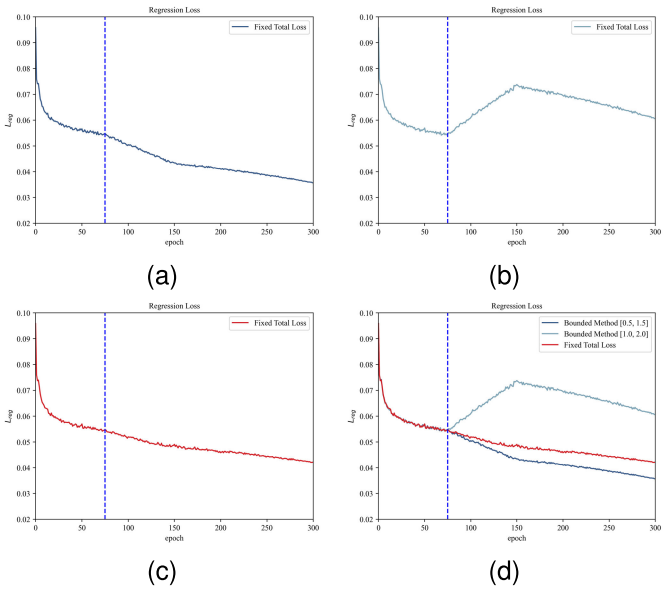



Fig. 9. Influence of different weighting rules on regression loss. (a) and (b) Use bounded method to determine the weight. Where the boundary interval of (a) is [0.5, 1.5] and the interval of (b) is [1.0, 2.0]. (c) Change of regression loss when the fixed total loss method is used to determine the weight. The blue dotted line is the node where our regression-guided refocusing learning works.

on the model performance is verified. These methods include fixed weights fluctuating between [0.5, 1.5], fixed weights fluctuating between [1.0, 2.0], and flexible weight adjustment methods with fixed total loss unchanged. The formula for calculating the weight of each method is shown in (15), (16),

TABLE III
COMPARISON OF DIFFERENT WEIGHTING RULES ON AI-TOD

| Baseline | Weighting algorithm | $AP$ | $AP_{50}$ | $AP_{75}$ |
|---|---|---|---|---|
| YOLOX-S | Bounded Method [0.5, 1.5] | 24.6 | 56.3 | 17.7 |
| | Bounded Method [1.0, 2.0] | 24.9 | 56.7 | **18.3** |
| | Fixed Total Loss | **25.0** | **57.0** | 18.1 |

TABLE IV
COMPARISON OF RRL COMBINED WITH DIFFERENT LOSS FUNCTIONS

| Baseline | $L_{reg}$ | $L_{cls}$ | $L_{pos}$ | $AP$ | $AP_{50}$ | $AP_{75}$ |
|---|---|---|---|---|---|---|
| YOLOX-S | | | | 20.6 | 49.4 | 13.8 |
| | ✓ | | | 24.8 | 56.7 | **18.1** |
| | ✓ | ✓ | | 24.9 | 56.9 | 18.0 |
| | ✓ | | ✓ | 24.7 | 56.2 | 17.8 |
| | ✓ | ✓ | ✓ | **25.0** | **57.0** | **18.1** |

and (3), where $\mathrm{IoU}_{\max}$ refers to the maximum IoU value in the IoU and $\mathrm{IoU}_{\min}$ indicates the minimum IoU value in the IoU

$$w_{g,p} = 1 + \alpha(t) \times \frac{\mathrm{IoU}_{g,p} - \mathrm{IoU}_{\mathrm{mean}}}{\mathrm{IoU}_{\max} - \mathrm{IoU}_{\min}} \tag{15}$$

$$w_{g,p} = 1 + \alpha(t) \times \frac{\mathrm{IoU}_{g,p} - \mathrm{IoU}_{\min}}{\mathrm{IoU}_{\max} - \mathrm{IoU}_{\min}}. \tag{16}$$

Related experiments were carried out on AI-TOD dataset. Fig. 9 shows the regression loss under different weighting rules. Fig. 9(a) and (b) uses bounded method to determine the weight, where (a) is determined by (15) and its boundary is [0.5, 1.5], (b) is determined by (16) and its boundary is [1.0, 2.0]. Fig. 9(c) is the change of regression loss when the weight is determined by (3). Obviously, compared with the value of constraint weight in a certain interval, the dynamic weight of fixed total loss can make the loss decline smoother and the training more stable. According to the experimental results shown in Table III, the dynamic weight of fixed total loss achieves better tiny object detection performance. Therefore, our subsequent experiments will all be based on the third weight calculation method.

*2) RRL Combined With Different Loss Functions:* The efficacy of RRL applied across various loss functions for improving tiny object detection performance has been substantiated. Ablation experiments were carried out on AI-TOD dataset, and detailed experimental results are presented in Table IV. RRL represents an optimization strategy specifically devised to refine model regression parameters. By assigning distinct weights to positive samples and adjusting the optimization sequence, it effectively unleashes the potential of the baseline model to achieve precise localization, thereby augmenting the accuracy of tiny object detection. Furthermore, the impact of incorporating RRL into both object classification loss and object positioning confidence loss was examined. The experimental results indicate that simultaneously integrating RRL into these loss functions further enhances the baseline model's tiny object detection performance compared to solely using RRL in regression loss. Based on meticulous experimental data, we uniformly apply RRL optimization across regression loss, classification loss, and object positioning confidence loss.

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TABLE V
EFFECTIVENESS OF DIFFERENT COMPONENTS ON AI-TOD

| Method | $CAFR$ | $RRL$ | $AP$ | $AP_{50}$ | $AP_{75}$ | Params | FPS |
|---|---|---|---|---|---|---|---|
| YOLOX-S | | | 20.6 | 49.4 | 13.8 | **8.94** | **36.7** |
| | ✓ | | 24.4 | 55.2 | 18.4 | 11.32 | 34.3 |
| | | ✓ | 25.0 | 57.0 | 18.1 | **8.94** | **36.7** |
| **CAF$^2$Net-S** | ✓ | ✓ | **28.1** | **61.4** | **21.8** | 11.32 | 34.3 |

*3) Evaluation of Different Components:* The effectiveness of the CAFR module and the regression-guided refocusing learning was independently validated on AI-TOD dataset. Compared to the baseline model, incorporating the CAFR module in the feature fusion neck network improves the $AP_{50}$ metric by 5.8%. Without altering the YOLOX-S structure, optimizing model parameters using the regression-guided refocusing learning increases the $AP_{50}$ metric by 7.6%. Building upon this, the combined usage of the CAFR module and the regression-guided refocusing learning achieves superior performance in detecting tiny objects compared to using either method alone, with the $AP_{50}$ metric surpassing the baseline model by 12.0%, reaching 61.4%. Besides, compared to the baseline model, our optimizations to the model architecture do not significantly impact the inference speed of the model. The effectiveness of our approach is thoroughly demonstrated, and Table V presents our experimental results.

## V. CONCLUSION

This article provides an in-depth analysis of the phenomenon that IoU metrics for tiny objects are highly sensitive to positional deviations and the feature misalignment problem introduced by deep feature maps during upsampling. To achieve more precise feature aggregation and accurate detection of tiny objects, CAF$^2$ENet, a CAF$^2$ENet, was proposed. In which, a CAFR module is include for multiscale features alignment. CAF$^2$ENet utilized the regression-guided refocusing learning for parameter tuning. Our method effectively tackles feature misalignment issues arising from upsampling, ensuring multiscale feature alignment, and reducing interference caused by complex backgrounds and densely distributed objects in tiny object detection. Simultaneously, the regression-guided refocusing learning thoroughly explores the potential for precise object localization in tiny object detection networks, resulting in superior performance for tiny object detection. Experimental results on the AI-TOD dataset showcase our method's superiority over other SOTA detection techniques, demonstrating advanced detection accuracy. Our method exhibits significant advantages when tackling tasks that demand high localization precision, such as detecting objects with limited pixels or detecting rotated objects. Further research in this area will be pursued in the future.

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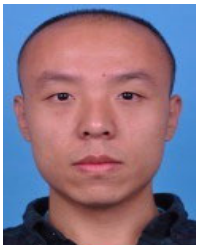
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